

COMPUTER NETWORKS

Week-3

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Address

- In order for a process running on one host to send packets to a process running on another host
- → the receiving process needs to have an address

In Internet, the host is identified by its IP address

Why IP address?

- Human beings can be identified by
 - names
 - CNIC numbers
 - Driver's license numbers
- Although each can be used to identify people, within a given context one identifier may be more appropriate than another
- E.g. we don't call each other by CNIC

IP address

- Internet hosts can also be identified in many ways like :
- Its **hostname** —such as www.facebook.com,
www.google.com
- However, hostnames provide little, if any, information about the location within the Internet of the host
- A hostname such as www.eurecom.fr → tells us that the host is probably in France
- Also note that hostnames can consist of variable-length alphanumeric characters, they would be difficult to process by **routers**
- For these reasons, hosts are also identified by so-called **IP addresses**

IP Address

- It is a 32 bit long identifier
- It encodes a network number (network prefix) and a host number

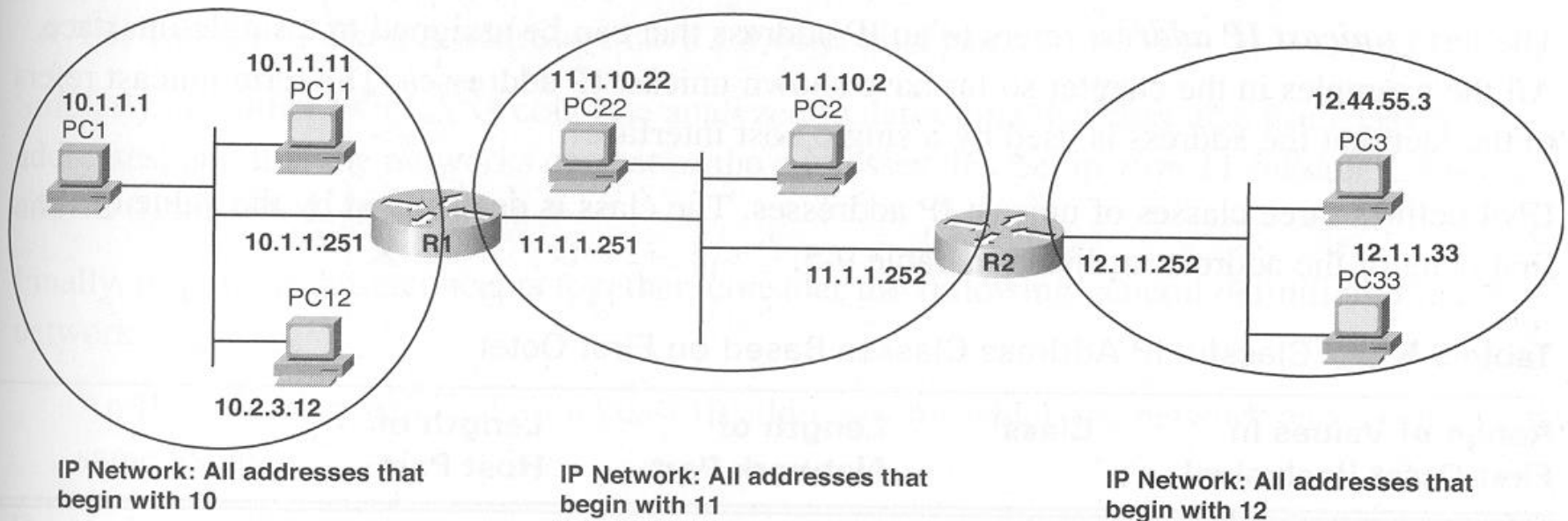
Address Assignment

- 3 key rules
 - Devices on the same LAN must use IP addresses in the same group (IP network)
 - Devices on different LANs that are separated by at least one **router** must use IP addresses in different IP networks
 - IP addresses must be unique inside the same internetwork

Internetwork

- A network could be defined as a group of locally connected computers (Eg: Computers connected in one school of BNU or in an office building)
- An internetwork is a collection of individual networks, connected by intermediate networking devices, that functions as a single large network

Figure 9-13 Three LANs, Three IP Networks



- Three networks

- All IP addresses that begin with 10
- All IP addresses that begin with 11
- All IP addresses that begin with 12

IP Address

- An *IP address* is a number that uniquely identifies every device on an IP network
- IP addresses operate at the Network layer of the TCP/IP protocol
- IP address must be unique and clients on the same network must have the **same** network ID

Two Flavors of IP Addresses

- **IPv4** or Internet Protocol version 4
- The Internet and most routers are currently configured for this
- Uses **32-bit** addresses
- About 290 million are reserved for special purposes
- A new version of IP addresses → **IPv6**
- Address size → **128-bit**
- To use IPv6 addresses, existing routers and hardware would need to be upgraded or configured to use this new version of IP addresses

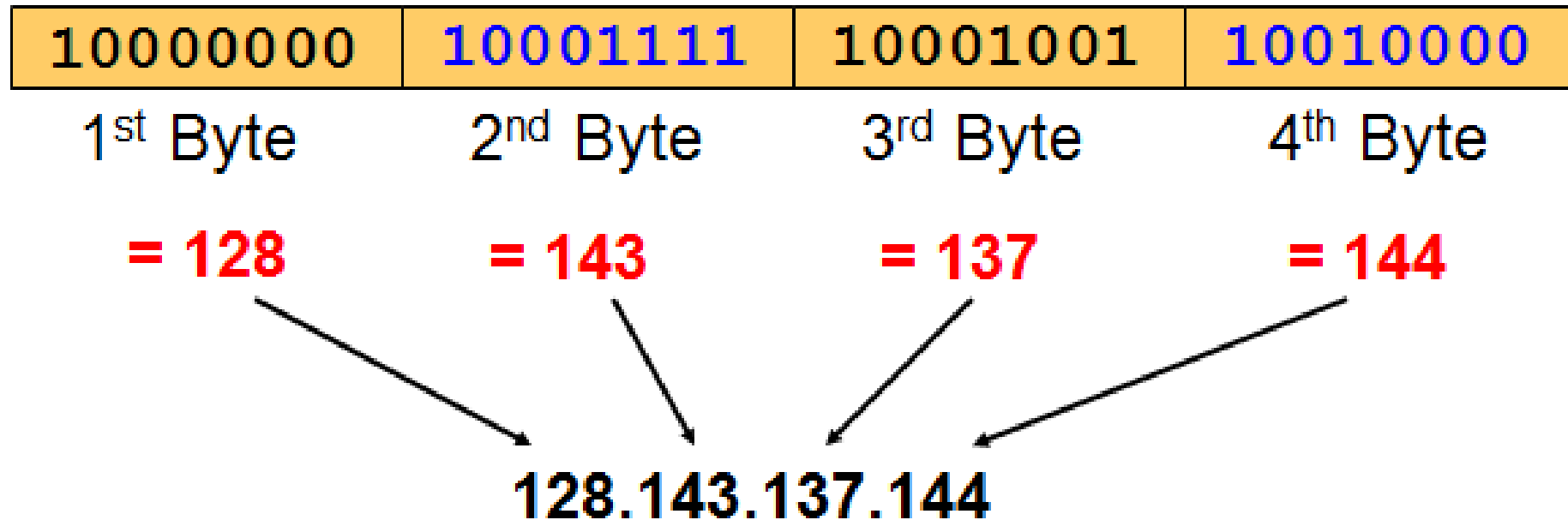
IPv4

- still commonly used but world is rapidly moving towards IPv6

Dotted-Decimal Notation

- IP addresses are usually represented in a format known as *dotted-decimal notation*
- In dotted-decimal notation, each group of eight bits — an *octet* — is represented by its decimal equivalent

Example



Example

11000000 10101000 10001000 00011100

- The dotted-decimal equivalent to this address is:
- 192.168.136.28

Example - Contd.

- 192 represents the first eight bits (11000000)
- 168 → the second set of eight bits (10101000)
- 136 → the third set of eight bits (10001000)
- 28 → the last set of eight bits (00011100)

IP Address

- the IP address is broken into two parts
- The network address
- The local/host address
- Each network has a unique network address
- Each device on a network has the same network address portion in its IP address
- The host address uniquely identifies a computer within a network
- Possible to have same host/local address on different networks
- The complete IP address (network address + host address) is always unique

IP Address

- Each physical network in the Internet is assigned a unique **network** number
- The network number appears as a prefix in the IP address
- No two networks on the Internet can be assigned the same network number
- No two computers on a given network can be assigned the same suffix/host address

IP address classes (IPv4)

- The IP protocol defines five different address classes:
- A, B, C, D, and E
- Each of the first three classes, A-C, uses a different size for the network ID and host ID portion of the address
- Class D is for a special type of address called a *multicast address*
- Class E is an *experimental* address class that isn't used

Classes of Addresses

- The first four bits of the IP address are used to determine into which class a particular address fits
 - ✓ If the first bit is 0 → Class A
 - ✓ If the first bit is 1 and the 2nd bit is 0 → Class B
 - ✓ If the 1st two bits are both 1 and the 3rd bit is 0 → Class C
 - ✓ If the first 3 bits are all 1 and the 4th bit is 0 → Class D
 - ✓ If the first 4 bits are all 1 → Class E

IP Address Classes

- For example, the designers may have decided that half of the address (16 bits) would be used for the network
- The remaining 16 bits would be used for the host ID
- The result of that scheme would be that the Internet could have a total of 65,536 (2^{16}) networks
- Each of those networks could have 65,536 hosts

Need to Classify?

- Suppose that a network of 1,000 computers joins the Internet and is assigned one of these hypothetical network IDs
- The network uses only 1,000 of its 65,536 host addresses
- More than 64,000 IP addresses would be wasted

Address Classes

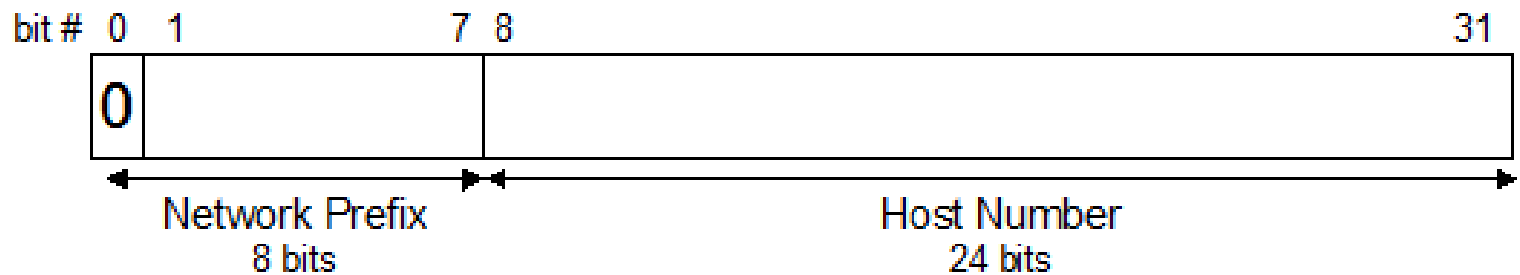
Class	Length of Network Part	Length of Host Part
A	1 octet	3 octets
B	2 octets	2 octets
C	3 octets	1 octet

Address Classes

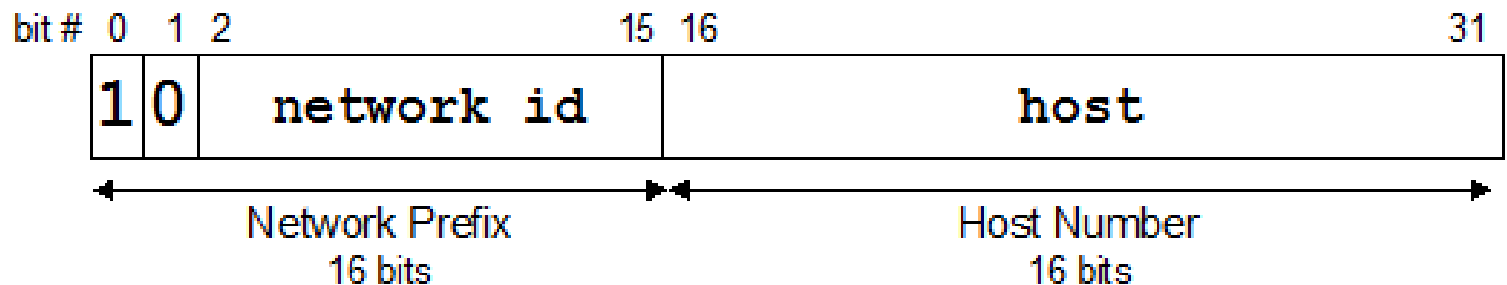
- 8 bits for the network leaves 24 bits for hosts. This gives us 256 networks each of which can have 16,777,216 hosts.
- 16 bits for the network leaves 16 bits for hosts. This gives us 65,536 networks each of which can have 65,536 hosts.
- 24 bits for the network leaves 8 bits for hosts. This gives us 16,777,216 networks each of which can have 256 hosts.

Classful IP Addresses

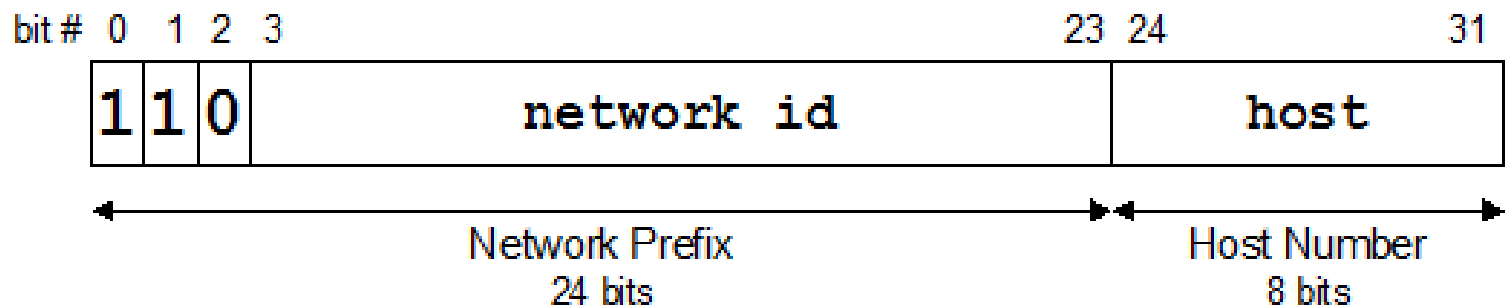
Class A



Class B

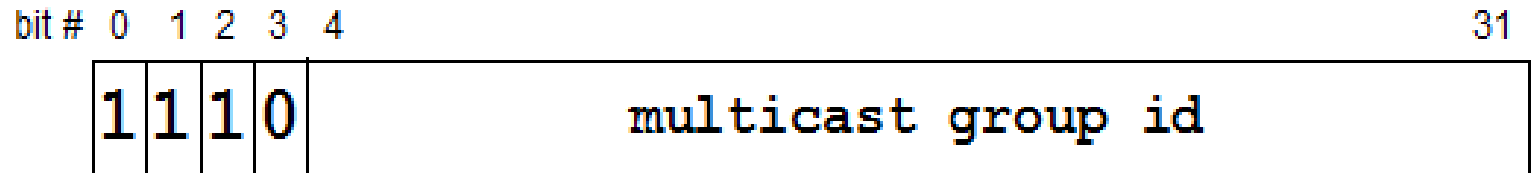


Class C

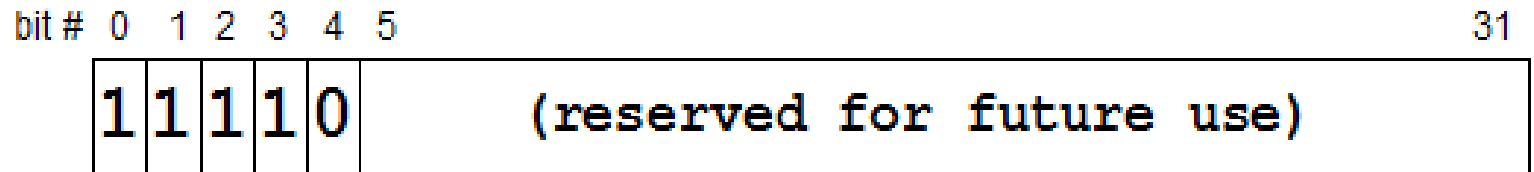


More Classes (TBDL)

Class D



Class E



Range of IP Addresses

- Class A addresses start with a binary 0 hence the first byte is in the range 0-127

7	6	5	4	3	2	1	0	Decimal Value
0	0	0	0	0	0	0	0	0
0	*	*	*	*	*	*	*	-
0	1	1	1	1	1	1	1	127

Class A From 0.0.0.0 to 127.255.255.255

Range of IP Addresses

- Class B addresses start with binary 10 hence the first byte is in the range 128-191

7	6	5	4	3	2	1	0		Decimal Value
1	0	0	0	0	0	0	0		128
1	0	*	*	*	*	*	*		-
1	0	1	1	1	1	1	1		191

Range of IP Addresses

- Class C addresses start with binary 110 hence the first byte is in the range 192-223

7	6	5	4	3	2	1	0		Decimal Value
1	1	0	0	0	0	0	0		192
1	1	0	*	*	*	*	*		-
1	1	0	1	1	1	1	1		223

Class A

- 8 bits for the network part and 24 for host part
- No. of possible hosts = 2^{24}
- Out of these 8 network bits, 1 bit is again fixed
- Leaving 7 bits to design networks
- $2^7 = 128$ possible network IDs
- E.g.
- 32.0.0.0 → Class A (first address in this network)
- 32.255.255.255 → last address in this network
- First IP address represents the network itself
- Last IP address represents the broadcast address (to send a packet to all hosts in this network) → directed broadcast address

Class A

- IP Address : 10.2.100.17
 - Default Class A subnet mask : 255.0.0.0
 - DBA : 10.255.255.255
-
- A **subnet mask** determines which portion of the IP address is the network and which is the host

Range of IP Addresses

- Class B addresses start with binary 10_2 hence the first byte is in the range 128-191

7	6	5	4	3	2	1	0		Decimal Value
1	0	0	0	0	0	0	0		128
1	0	*	*	*	*	*	*		-
1	0	1	1	1	1	1	1		191

Class B

- Class B can be diagnosed from first decimal value:
- $128 - 191 \rightarrow 64$ values
- $10\ 000000 = 128$
- $10\ 000001 = 129$
- $10\ 000010 = 130$
- $10\ 000011 = 131$
- .
- .
- $10\ 111111 = 191$

Class B

- Total bits in IP address = 32
- First 2 bits are fixed = $32 - 2 = 30$
- Total possible addresses in class B = 2^{30}
- Network ID = 2 octets
- 2 bits are fixed $\rightarrow$ 6 bits + 8 bits = 14 bits
- Host ID = 2 octets
- Possible networks = $2^{14} = 16384$
- Possible hosts = $2^{16} = 65536$ in each network
- Usable host IDs = $65536 - 2 = 65534$

Class B

- For a network starting with 128
- Networks will be : 128.0.0.0 - 128.0.255.255
- first IP address denotes the network itself
- last IP address is broadcast address
- remaining are $65536 - 2 = 65534$ hosts (usable in this network)

Class B

- 130.1.2.3 (given host IP address)
- class B from 130
- To find out the network address
- Default mask = 255.255.0.0 (from Class B)
- 130.1.2.3
- AND
- 255.255.0.0
-
- 130.1.0.0 (bcz AND with zero)
- **this is network address**
- so 130.1.2.3 belongs to 130.1.0.0 network
- if this address is for BNU then whole world knows you as **130.1.0.0**
- 130.1.0.0 ----- 130.1.255.255 : Possible IP addresses in this network
- Network Address → 130.1.0.0
- Directed broadcast Address → 130.1.255.255

Range of IP Addresses

- Class C addresses start with binary 110_2 hence the first byte is in the range 192-223

7	6	5	4	3	2	1	0		Decimal Value
1	1	0	0	0	0	0	0		192
1	1	0	*	*	*	*	*		-
1	1	0	1	1	1	1	1		223

Class C

- 192.168.10.11
- Network Address → 192.168.10.0
- default Class C subnet mask 255.255. 255.0
- DBA = 192.168.10.255

Remember for IPv4

- If all the bits in the host part are "0", that represents the network address
- If all the bits in the host part are "0" except the last bit, it is the first usable IP address
- If all the bits in the host part are "1" except the last bit, it is the last usable IP address
- If all the bits in the host part are "1", that represents the DBA

For IPv4

IP Class	First Octet Range	Default Subnet Mask	Number of Blocks	Number of IP Addresses Per Block
A	1 - 126	255.0.0.0	126	16,777,214
B	128 - 191	255.255.0.0	16,384	65,534
C	192 - 223	255.255.255.0	2,097,152	254

Class Exercise

- Write an IP address that belongs to class B
- Write the default Subnet Mask
- Write the Network Address

Sample solution

TASK	SOLUTION-Sample
Write an IP Address that belongs to Class-B	172.16.189.5 (128-191 → Class B)
Write the default Subnet Mask	255.255.0.0
Write the Network Address	172.16.0.0 (Last 2 octets in above address → zero)

Book-B

- Chapter 21
- Douglas E. Comer